2110201: COMP ENG MATH (2023/2) Quiz 1

Name: Student ID: Entry Number:

1. (8 points) Given a function $f\colon \mathbf{R}^4 \Rightarrow \mathbf{R}^3$ such that

$$f(a, b, c, d) = (a + b + 2d, 2a + 3b - c + 7d, b - c + 3d)$$

Let $\mathbf{A}$ be a matrix that represents the linear transformation f.

1.1. (2 points) Find the matrix $\mathbf{A}$.

1.2. (3 points) Find RRE of the matrix $\mathbf{A}$.

1.3. (2 points) Find all vectors $\mathbf{x}$ such that $\mathbf{Ax} = \mathbf{0}$

1.4. (0.5 points) Is the function f one-to-one? (Answer **Yes**/**No** only)

1.5. (0.5 points) Is the function f onto? (Answer **Yes**/**No** only)

2. (2 points) Answer **True** or **False** for each sub question in the table provided.

2.1	**2.2**	**2.3**	**2.4**

2.1. $G(x, y, z) = (x^2, y)$ is a linear transformation.

2.2. Given matrix $\mathbf{T} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $\mathbf{x} = (1, 1)$, and $\mathbf{b} = \mathbf{Tx} = (3, 7)$. We can conclude that (1, 1) is a pre-image of (3, 7) respect to $\mathbf{T}$.

2.3. The vector (1, 1) is an eigenvector of the matrix $\begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix}$.

2.4. Please answer **True** for a free point.